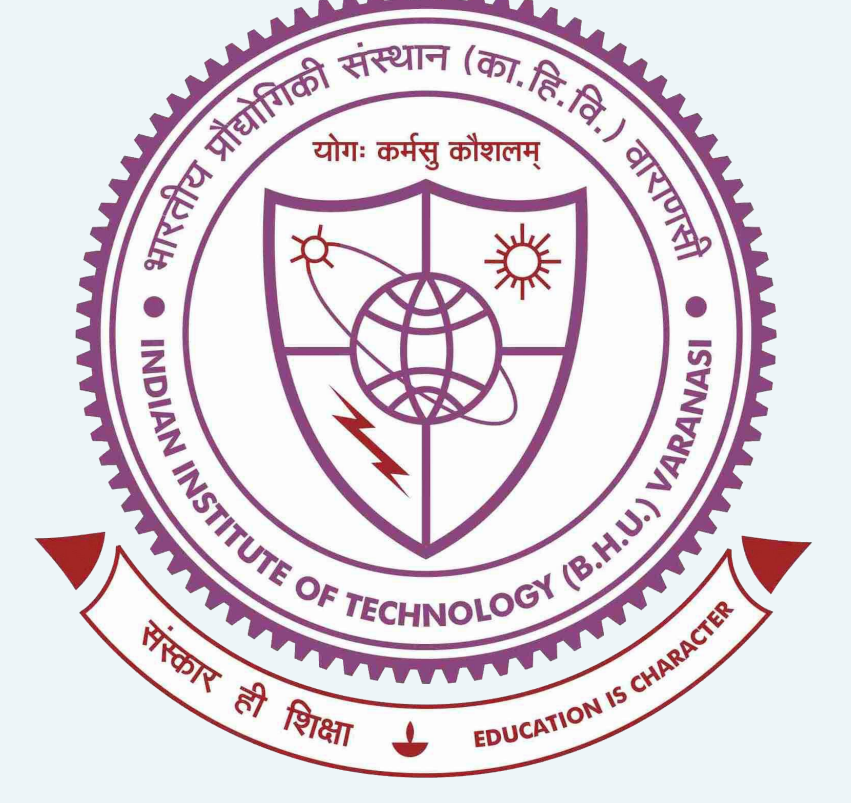
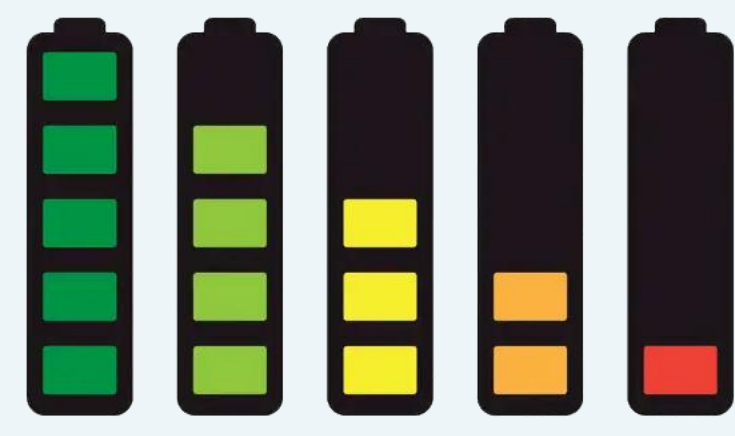


Physics-based Predictive Modelling to Decipher Phase Transitions and Gas Evolution in High-energy NMC811 Electrodes



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Overview

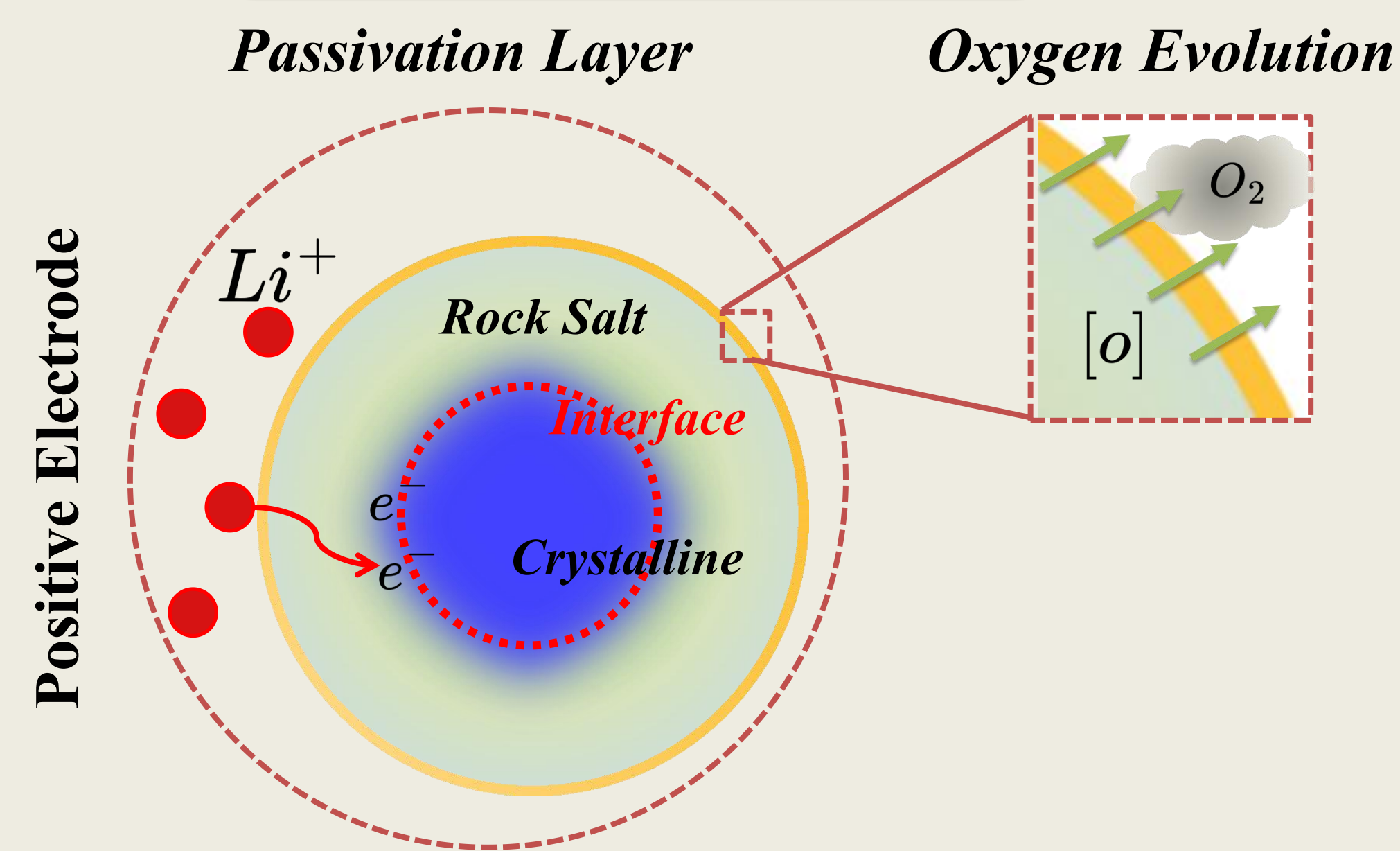
Introduction

- ❖ **Novelty:** Thermally-Coupled @ CC-CV
- ❖ **Utility:** degradation-Integrated
- ❖ **Background:** Single Particle Model

Problem Statement:-

Ni-rich layered cathodes (eg., NMC811) offer high specific capacities but suffer from severe degradation at high voltage due to **oxygen evolution**, **structural reconstruction**, and **passivation layer growth**. These degradation mechanisms induce **loss of active material (LAM)**, **loss of lithium inventory (LLI)**, and **thermal instability**, leading to capacity and power fade over repeated cycling.

Degradations



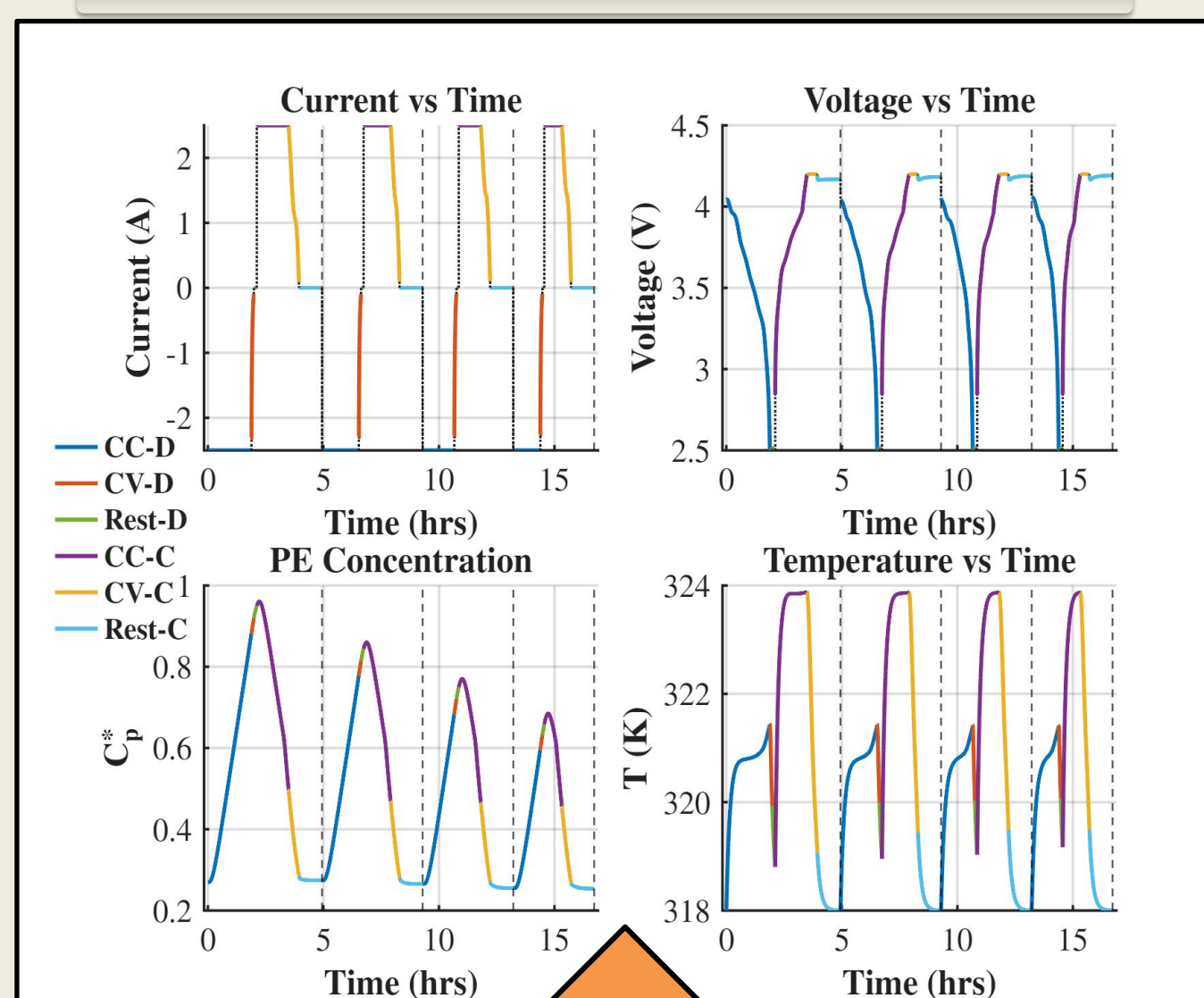
References

Publications

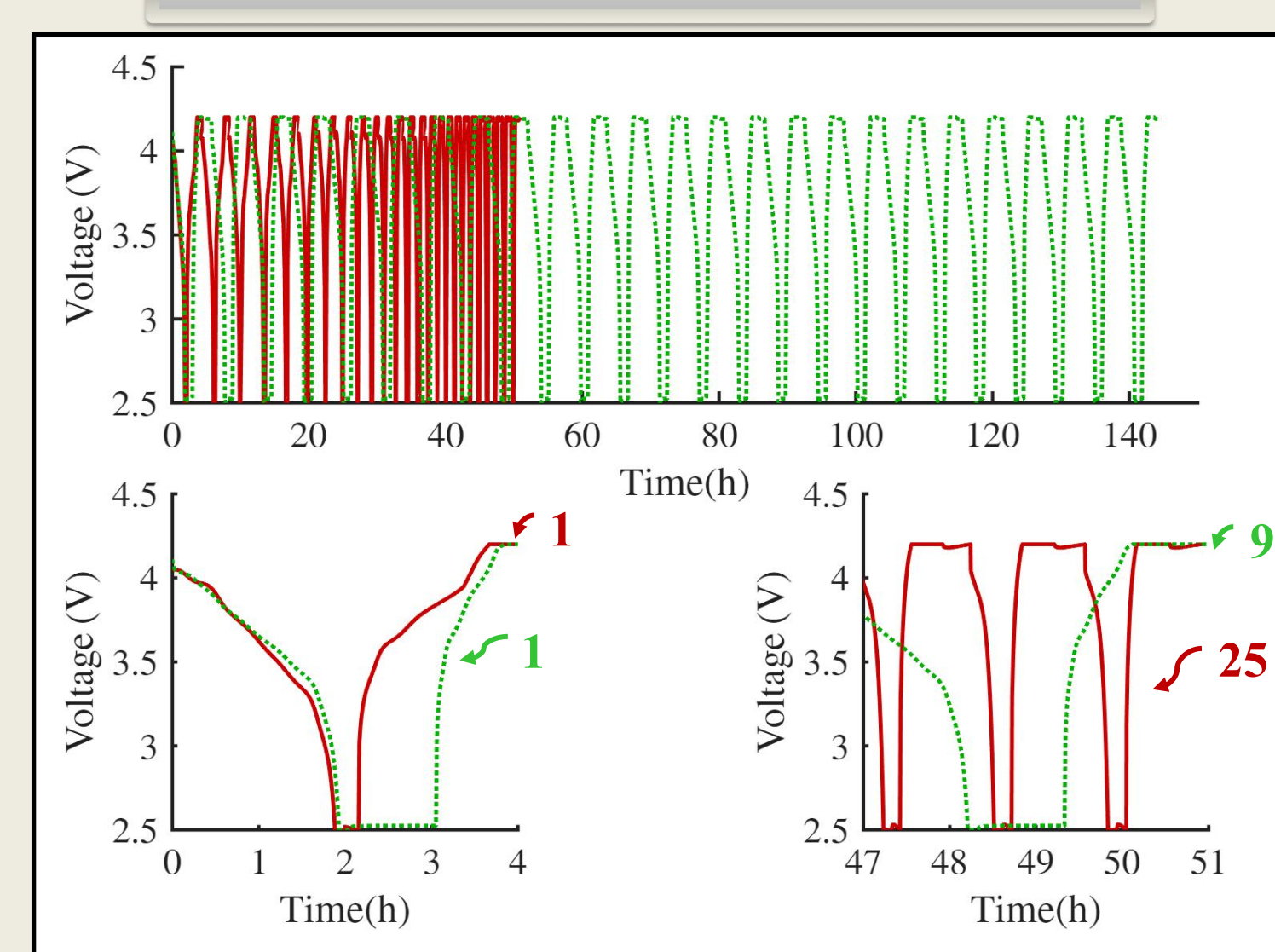
- [1] Ghosh, A., et al. J. Electrochem. Soc., **2021**, 168, 020509.
- [2] Chen, C., et al. J. Electrochem. Soc., **2020**, 167, 080534.
- [3] Guo, M., et al. J. Electrochem. Soc., **2011**, 158, A122.
- [4] Zhuo, M., et al. J. Power Sources. **2023**, 556, 232461.
- [5] O'Regan, K., et al. Electrochim. Acta, **2022**, 425, 140700.

Results

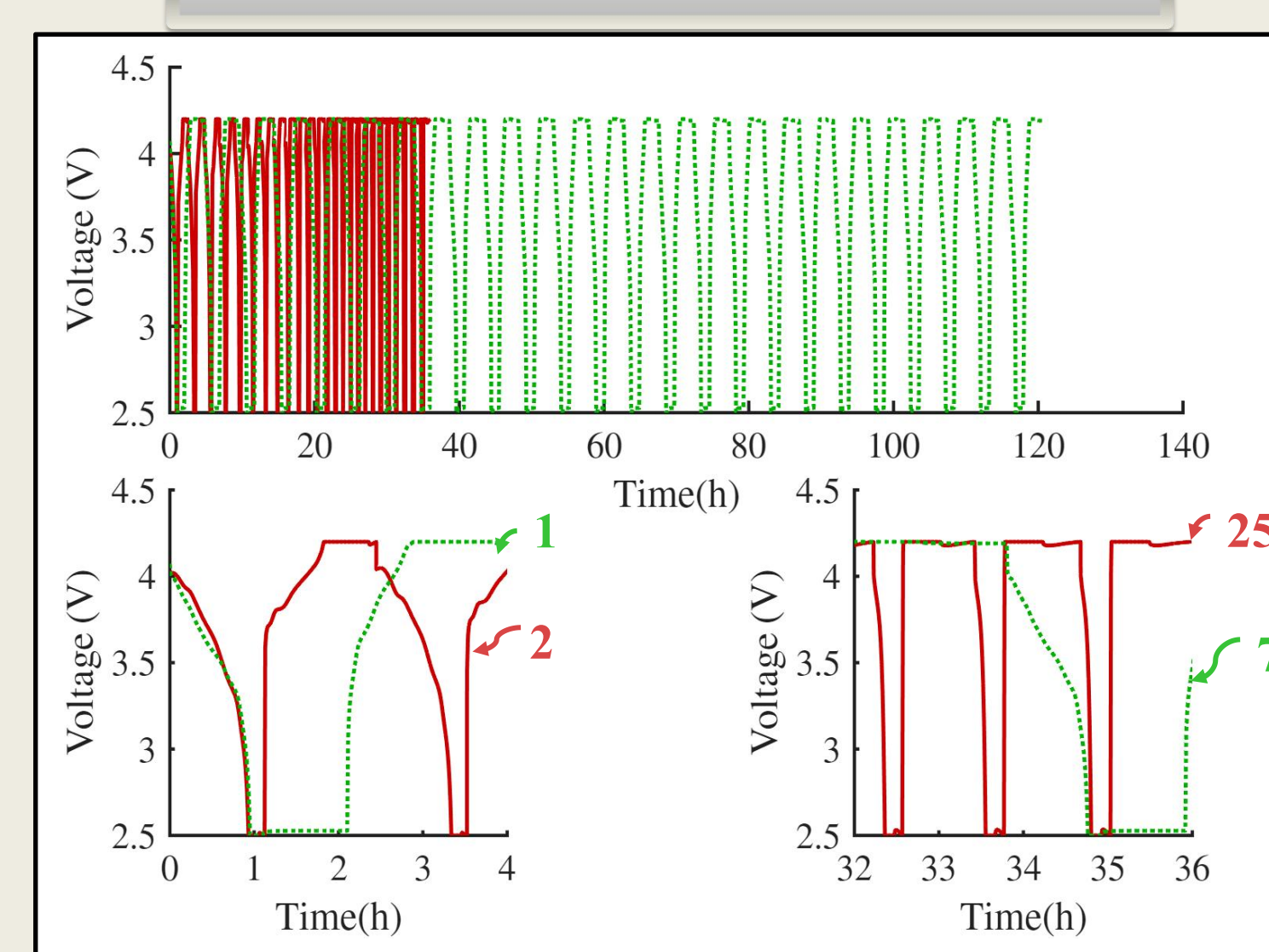
CC-CV-T-dSPM



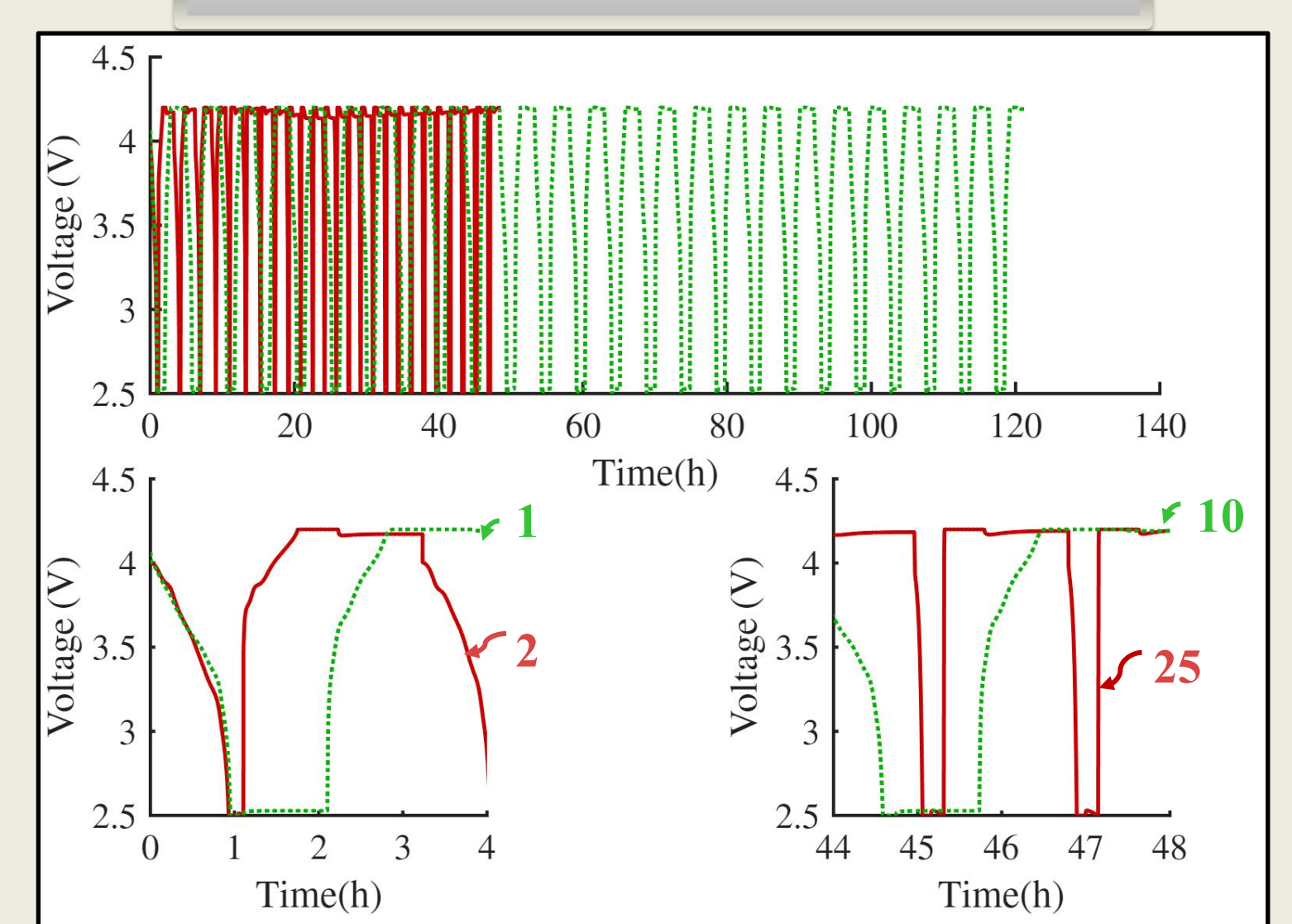
0.5C and 25°C



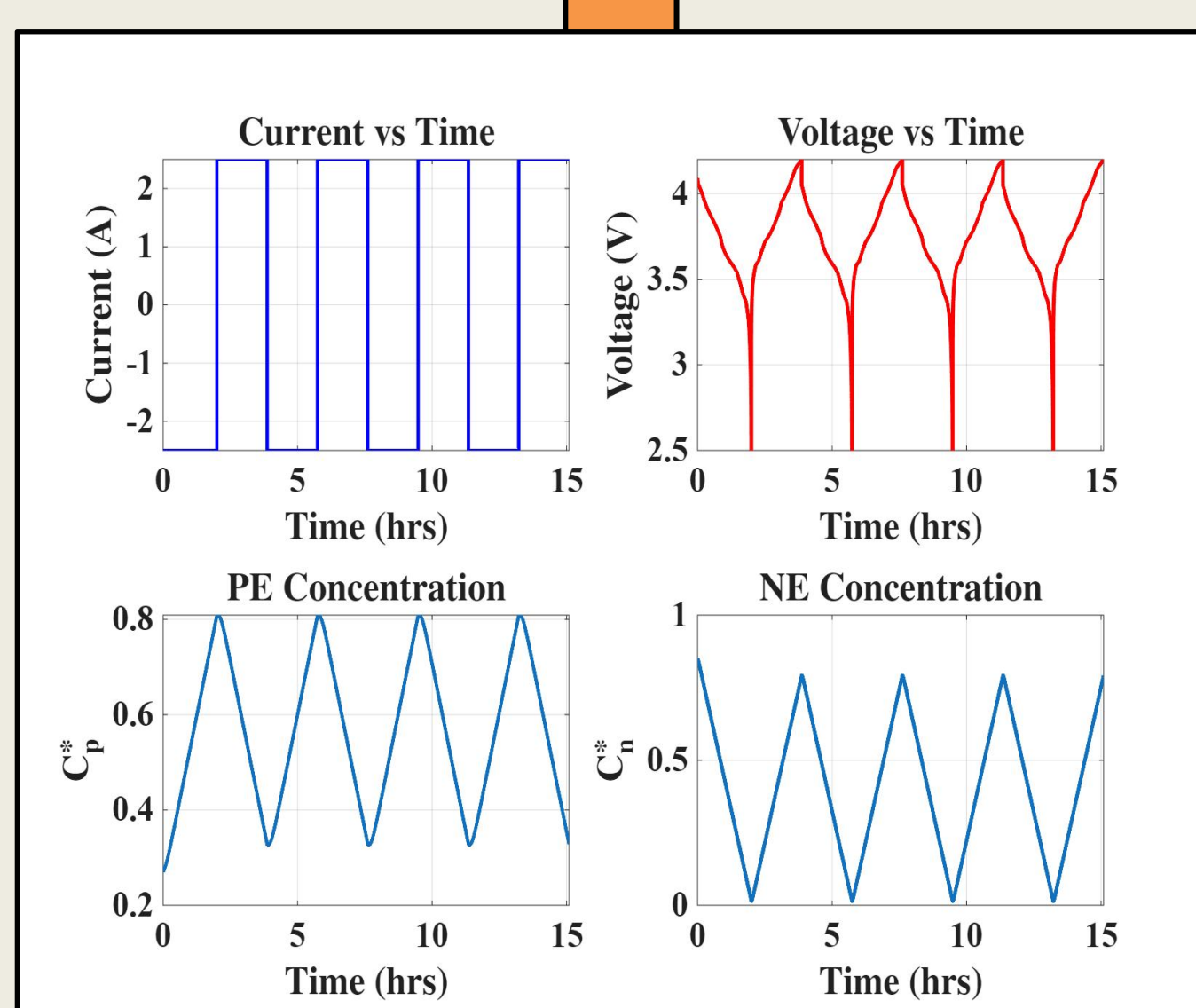
1C and 25°C



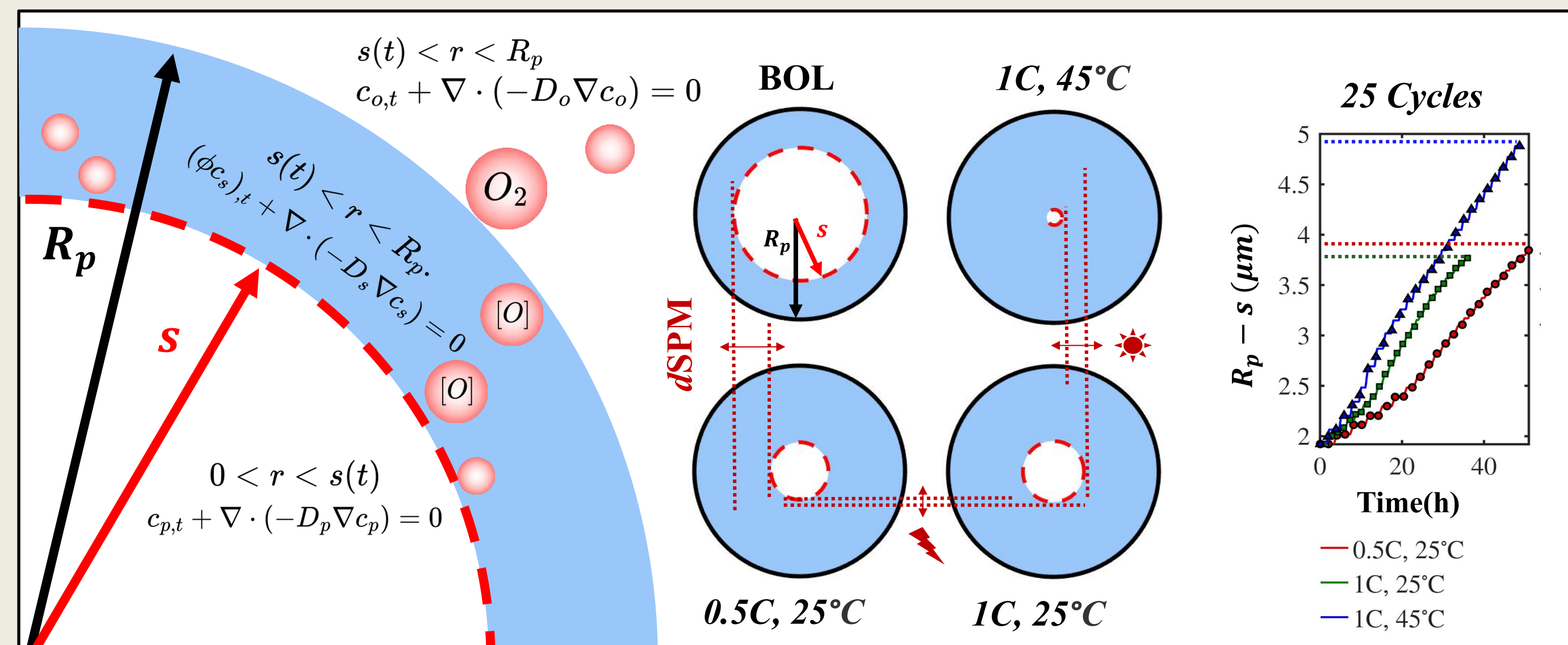
1C and 45°C



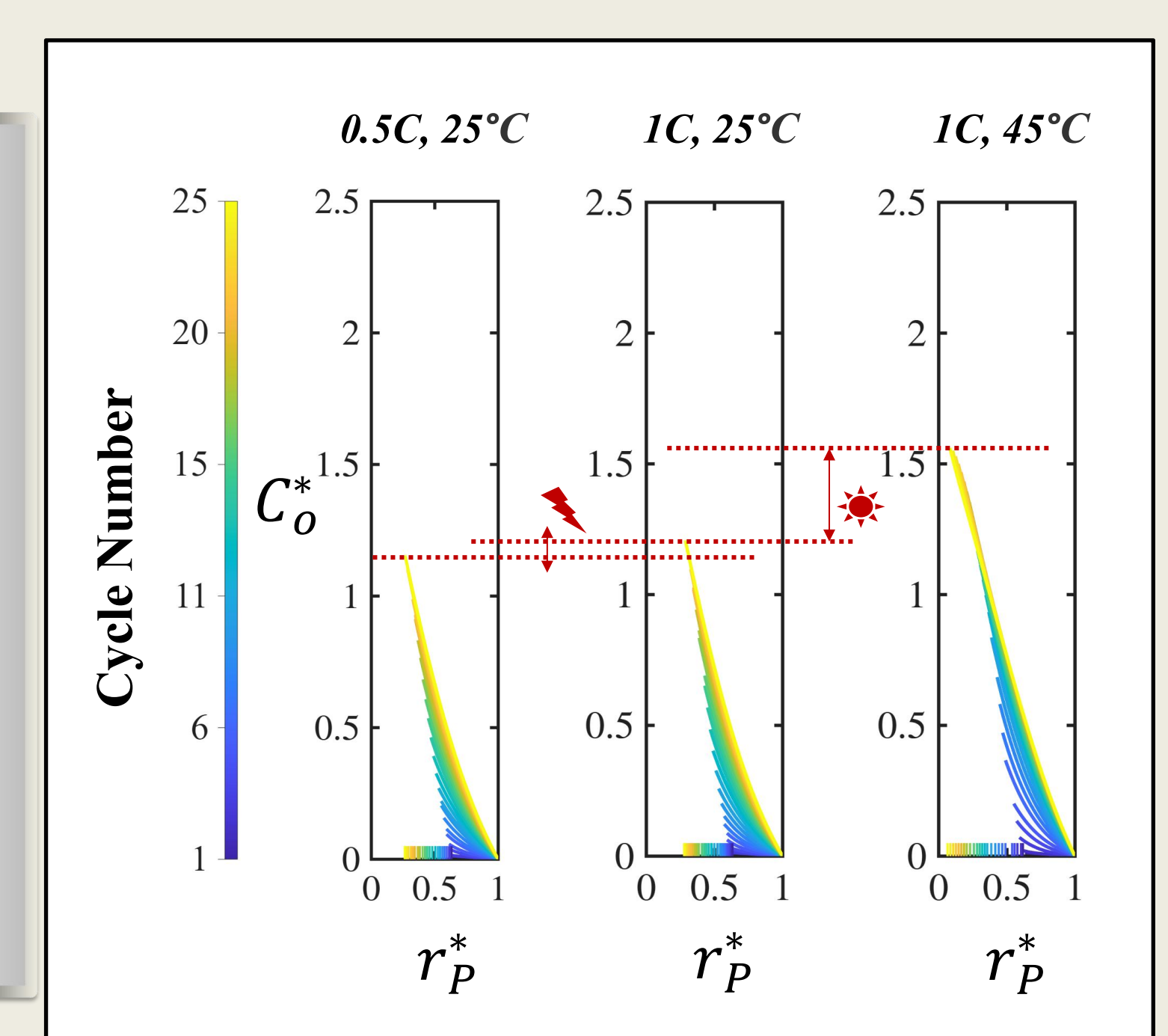
CC SPM



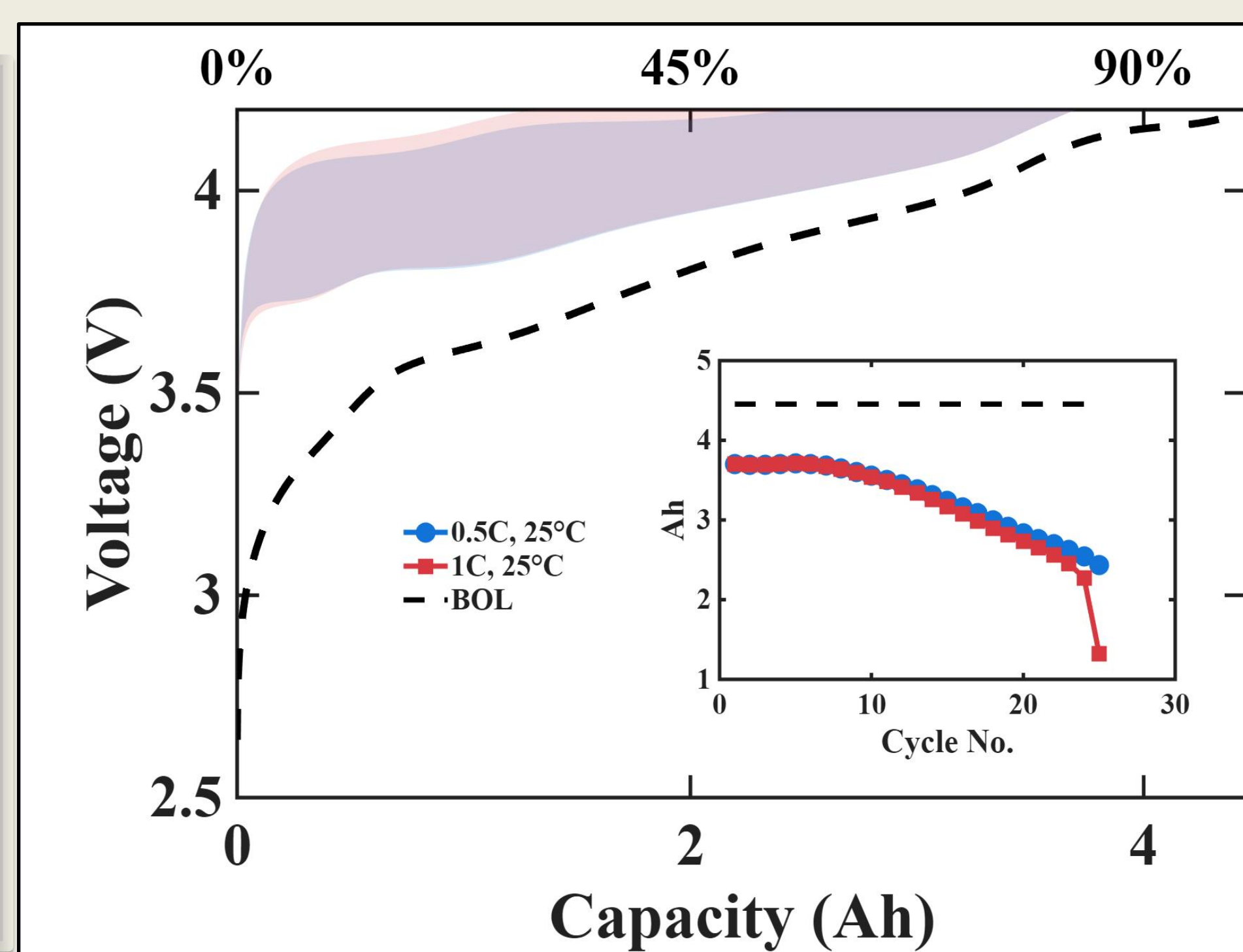
Shrinking Core



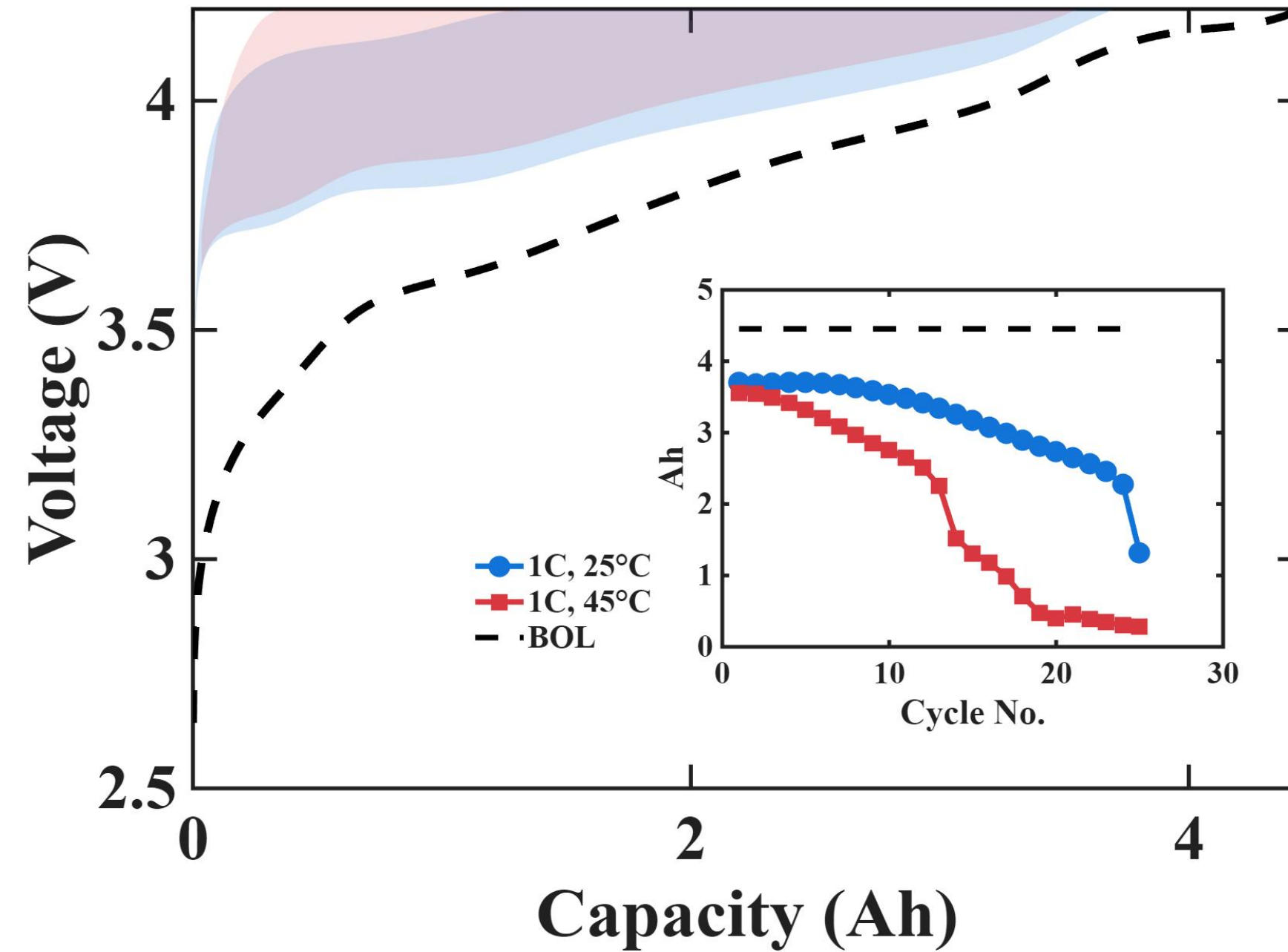
O₂ Evolution



0.5C and 1C at 25°C



Capacity (Ah)

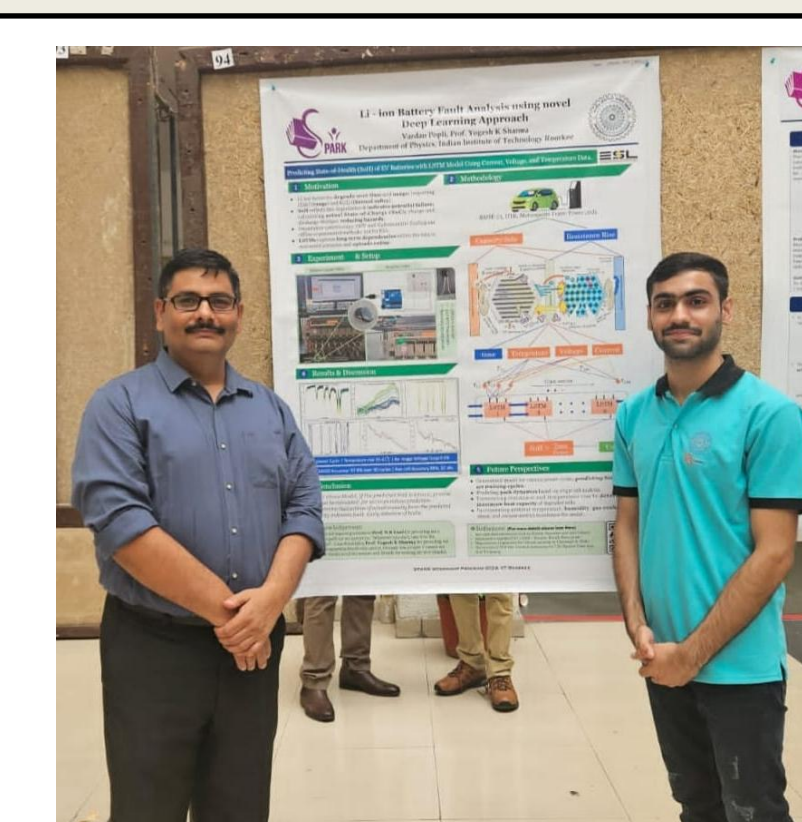
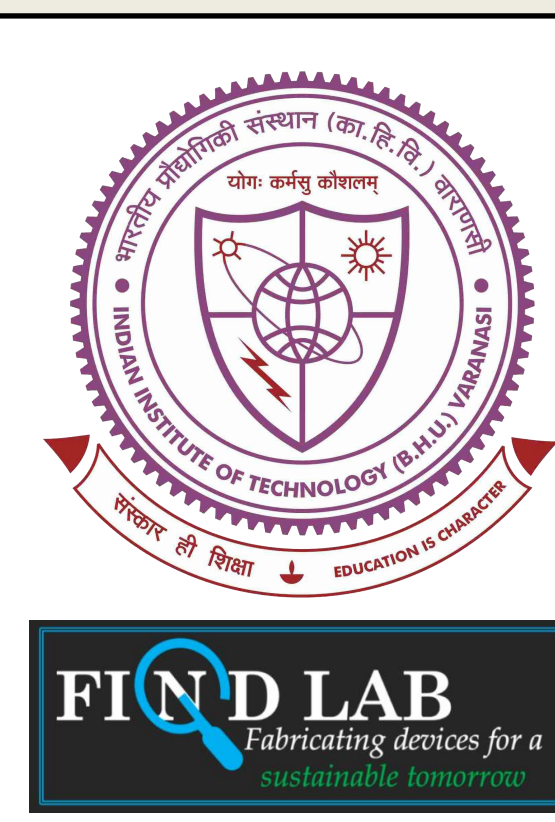
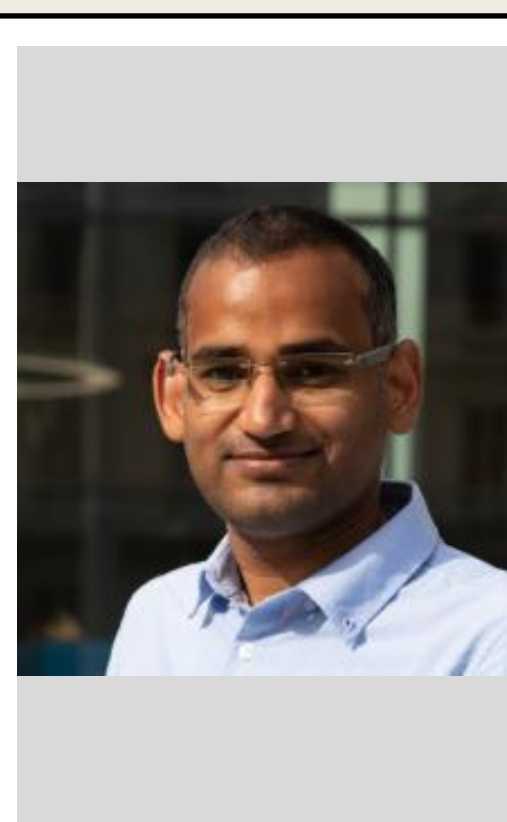
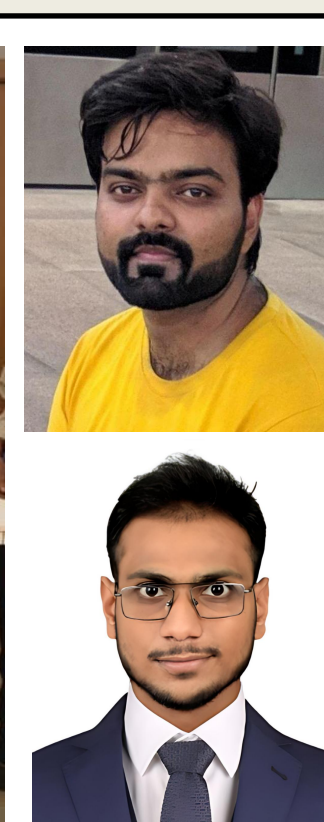


1C at 25°C and 45°C

Conclusion

INITIAL STABLE OPERATION		
NMC Crystalline	Balanced Li ⁺ Transport	Intact Electrode Structure
ONSET OF DEGRADATION		
Loss of Active Material	Passivation Growth	Loss of Lithium Inventory
ACCELERATED VOLTAGE DROP/ FAILURE		
Severe Impedance Growth	Structural Breakdown	Thermal Instability

Thanks



Father



Behind the scene